

一、(1) 写出氢原子的能级, 以及每个能级对应的简并度;

(2) 简要说明原子中存在“精细结构”, “超精细结构”的原因;

(3) GaAs 是一类重要的半导体, 其电子结构与氢原子类似, 已知

GaAs 的相对介电常数 $\epsilon_r = 3$, 相对原子质量 $m^* = 1.2m$, 玻尔半径

$a = 0.07a_0$ (注: 氢原子的相对原子质量 $m = 1.00794$, 玻尔半径

$a_0 = 52.9 \text{ pm}$), 求 GaAs 的电子结构的能级.

$$(1) E_n = -\frac{e^2}{2an^2} \quad \text{简并度 } 2n^2$$

(2) 精细结构: 自旋-轨道耦合

超: 偶极矩之间的电磁相互作用

二、一维空间中有一势场 $V_0 = \begin{cases} A\delta(x-a) \\ A\delta(x+a) \end{cases}$, 已知偶宇称时空间波函数

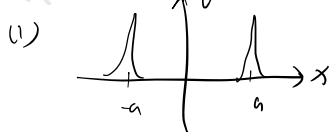
$$\psi(x) = \begin{cases} Be^{k_1x}, & x \leq -a \\ \cosh k_1x, & -a < x < a \\ Be^{-k_1x}, & x \geq a \end{cases} \quad \text{奇宇称时 } \psi(x) =$$

$$\begin{cases} Ce^{k_2x}, & x \leq -a \\ \sinh k_2x, & -a < x < a \\ -Ce^{-k_2x}, & x \geq a \end{cases}$$

(1) 写出波函数所满足的连续性条件;

(2) 利用连续性条件, 说明 k_1 与 B 所满足的关系或 k_2 与 C 所满足的关系;

(3) 证明奇宇称时满足关系;



$$\psi(a^+) = \psi(a^-)$$

$$\psi(-a^+) = \psi(-a^-)$$

$$\psi(a^+) - \psi(a^-) = \frac{2mA}{\hbar^2} \psi(a)$$

$$\psi(-a^+) - \psi(-a^-) = \frac{2mA}{\hbar^2} \psi(-a)$$

(2) 偶宇称

$$\psi(x) = \begin{cases} Be^{k_1x}, & x \leq -a \\ \cosh k_1x, & -a < x < a \\ Be^{-k_1x}, & x \geq a \end{cases}$$

$$Be^{-k_1a} = \cosh k_1a$$

$$-Bk_1e^{-k_1a} - (k_1 \sinh k_1a) = \frac{2mA}{\hbar^2} Be^{-k_1a}$$

$$\sinh k_1a = \frac{e^{k_1a} - e^{-k_1a}}{2}$$

$$-Bk_1e^{-k_1a} - \frac{1}{2}k_1e^{k_1a} + \frac{1}{2}k_1e^{-k_1a} = \frac{2mA}{\hbar^2} Be^{-k_1a}$$

$$k_1 \left(\frac{1}{2} - B \right) - \frac{1}{2}k_1e^{2k_1a} = \frac{2mA}{\hbar^2} B$$

三、已知一个自旋为 1 的体系, $H = AS_z + BS_x^2$.

(1) 写出此哈密顿量对应的能量本征值;

(2) 假设初始时体系处于 S_z 的本征值为 $\hbar$ 的本征态, 求当 $t > 0$ 时, 测得体系处于 S_z 的本征值为 $-\hbar$ 的概率.

$$\begin{aligned}
 (1) \quad \hat{H} &= A\hbar \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix} + \frac{B\hbar^2}{2} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \\
 &= \begin{pmatrix} A\hbar + \frac{B\hbar^2}{2} & 0 & \frac{B\hbar^2}{2} \\ 0 & B\hbar^2 & 0 \\ \frac{B\hbar^2}{2} & 0 & \frac{B\hbar^2}{2} - A\hbar \end{pmatrix}
 \end{aligned}$$

$$\text{解 } \hat{H}\psi = E\psi$$

$$\text{得 } E_1 = \frac{B\hbar^2}{2} + \hbar\sqrt{A^2 + \frac{B^2}{4}}$$

$$E_2 = B\hbar^2$$

$$E_3 = \frac{B\hbar^2}{2} - \hbar\sqrt{A^2 + \frac{B^2}{4}}$$

$$(2) \quad t=0 \text{ 时, 处于 } \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = |\psi(0)\rangle$$

$$S_z \text{ 为 } -\hbar \text{ 的本征态 } \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{令 } A\hbar = a, \quad \frac{B\hbar^2}{2} = b, \quad \sqrt{a^2 + b^2} = c$$

$$E_1 = b + c, \quad E_2 = 2b, \quad E_3 = b - c$$

$$\psi_1 = \frac{1}{\sqrt{b^2 + (a+c)^2}} \begin{pmatrix} c+a \\ 0 \\ b \end{pmatrix}$$

$$\psi_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\psi_3 = \frac{1}{\sqrt{b^2 + (a-c)^2}} \begin{pmatrix} a-c \\ 0 \\ b \end{pmatrix}$$

$$|\psi(t)\rangle = c_1 e^{-\frac{iE_1 t}{\hbar}} |\psi_1\rangle + c_2 e^{-\frac{iE_2 t}{\hbar}} |\psi_2\rangle + c_3 e^{-\frac{iE_3 t}{\hbar}} |\psi_3\rangle$$

$$t=0 \text{ 时 } |\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$c_1 = \langle \psi_1 | \psi(0) \rangle = 0$$

$$c_2 = \langle \psi_2 | \psi(0) \rangle = \frac{a+c}{\sqrt{b^2(a+c)^2}}$$

$$c_3 = \langle \psi_3 | \psi(0) \rangle = \frac{a-c}{\sqrt{b^2(a-c)^2}}$$

处于 $\psi = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ 概率

$$P = |\langle \psi | \psi(t) \rangle|^2 = \frac{a-c}{b} + \frac{b^2}{2c^2} \cos \frac{2ct}{\hbar}$$

四、已知一维谐振子体系的波函数满足 $\psi(x) = A e^{-\frac{\alpha^2 x^2}{2}}$, 其中 A 为归一化系数.

(1) 求系数 A , 并写出此谐振子的概率分布函数;

(2) 分别求出 $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$, $\langle p^2 \rangle$;

(3) 求出能量表象下此波函数的形式;

(4) 证明 $\langle \Delta x^2 \rangle \langle \Delta p^2 \rangle \geq \frac{\hbar^2}{4}$.

$$(1) \int_{-\infty}^{+\infty} |A e^{-\frac{\alpha^2 x^2}{2}}|^2 dx = 1$$

$$|A|^2 \cdot \sqrt{\frac{\pi}{\alpha^2}} = 1$$

$$|A|^2 = \sqrt{\frac{\alpha^2}{\pi}}$$

$$A = \sqrt{\frac{\alpha}{\sqrt{\pi}}}$$

概率分布

$$|\psi(x)|^2 = \frac{\alpha}{\sqrt{\pi}} e^{-\alpha^2 x^2}$$

$$(3) \psi(x) = \sqrt{\frac{\alpha}{\sqrt{\pi}}} e^{-\frac{\alpha^2 x^2}{2}}$$

$$\hat{p} = -\frac{\hbar}{i} \frac{d}{dx} + \frac{1}{2} m \omega^2 x^2$$

$$E = (n + \frac{1}{2}) \hbar \omega$$

$$(2) \langle x \rangle = \int_{-\infty}^{+\infty} \psi^*(x) \cdot x \cdot \psi(x) dx$$

$$= 0$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} (\psi(x))^* \cdot x^2 \cdot \psi(x) dx$$

$$= \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} x^2 e^{-\alpha^2 x^2} dx$$

$$= \frac{\alpha}{\sqrt{\pi}} \cdot \frac{1}{2\alpha^2} \cdot \sqrt{\frac{\pi}{\alpha^2}}$$

$$= \frac{1}{2\alpha^2}$$

$$\langle \hat{p} \rangle = \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-\frac{\alpha^2 x^2}{2}} \left(\frac{\hbar}{i} \frac{d}{dx} \right) e^{-\frac{\alpha^2 x^2}{2}} dx$$

$$= \frac{i\hbar\alpha^3}{\sqrt{\pi}} \int_{-\infty}^{+\infty} x e^{-\alpha^2 x^2} dx$$

$$(4) \langle \Delta x^2 \rangle = \overline{x^2} - (\bar{x})^2$$

$$= \frac{1}{2\alpha^2}$$

$$\langle \Delta p^2 \rangle = \frac{1}{2} \alpha^2 \hbar^2$$

$$\langle \Delta x^2 \rangle \cdot \langle \Delta p^2 \rangle = \frac{\hbar^2}{4} \geq \frac{\hbar^2}{4}$$

$$= \frac{i\hbar\alpha^3}{2\pi} \int_{-\infty}^{+\infty} \frac{1}{y} e^{-\alpha^2 x^2} dx^2$$

$$= \frac{i\hbar\alpha^3}{2\pi} \left(-\frac{e^{-\alpha^2 x^2}}{\alpha^2} \right) \Big|_{-\infty}^{+\infty}$$

$$= 0$$

$$\langle \hat{p}^2 \rangle = \int_{-\infty}^{+\infty} \psi^*(x) \left(-\hbar^2 \frac{d^2}{dx^2} \right) \psi(x) dx$$

$$= \frac{1}{2} \alpha^2 \hbar^2$$

II.

$$\hat{H} = \frac{1}{2m} \hat{p}^2 + \frac{1}{2} m \omega^2 \hat{x}^2$$

$$\hat{a}^+ = \frac{1}{\sqrt{2m\hbar\omega}} (i\hat{p} + m\omega\hat{x})$$

$$\hat{a}^- = \frac{1}{\sqrt{2m\hbar\omega}} (-i\hat{p} + m\omega\hat{x})$$

$$\begin{cases} \hat{p} = i\sqrt{m\hbar\omega} (\hat{a}^+ - \hat{a}^-) \\ \hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}^+ + \hat{a}^-) \end{cases}$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}^+ + \hat{a}^-)$$

六、已知一哈密顿量 H 的本征值为 E_1, E_2 相应的本征态为

$$|\psi_1\rangle, |\psi_2\rangle, \text{ 同时存在 } H' = \begin{pmatrix} \alpha & \beta_1 \\ \beta_2 & \alpha \end{pmatrix}$$

(1) 若 H' 可视为微扰, 利用微扰法求出体系的二级近似能量与一级近似波函数; E_2, E_1

(2) 若 $\beta_1 = \beta_2 = 0, E_1 = E_2$, 求出此时的一级修正能量与零级波函数.

$$(1) E_i' = E_i^{(0)} + E_i^{(1)} + E_i^{(2)}$$

$$= E_i + H'_{ii} + \frac{|H'_{2i}|^2}{E_i - E_2}$$

$$H'_{11} = (1 \ 0) \begin{pmatrix} \alpha & \beta_1 \\ \beta_2 & \alpha \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (\alpha \ \beta_1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \alpha$$

$$H'_{21} = (0 \ 1) \begin{pmatrix} \alpha & \beta_1 \\ \beta_2 & \alpha \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \beta_2$$

$$E_i' = E_i + \alpha + \frac{\beta_2^2}{E_i - E_2}$$

$$(2) H' = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}$$

$$E_1 = E_2, \varphi_1 = |\psi_1\rangle, \varphi_2 = |\psi_2\rangle$$

$$\psi^{(0)} = c_1 \varphi_1 + c_2 \varphi_2$$

$$\begin{pmatrix} H'_{11} & H'_{12} \\ H'_{21} & H'_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$H_{11} = \langle \psi_1 | H' | \psi_1 \rangle$$

$$= \alpha$$

$$H'_{12} = 0$$

$$H'_{21} = 0$$

$$H'_{22} = \alpha$$

$$(\alpha - E)^2 = 0$$

$$E = \alpha$$

简并仍未消除

$$E_2' = E_2 + H_{22}' + \frac{(H_{12}')^2}{E_2 - E_1}$$

$$H_{22}' = (0 \ 1) \begin{pmatrix} \alpha & \beta_1 \\ \beta_2 & \alpha \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \alpha$$

$$H_{12}' = \beta_1$$

$$E_1' = E_1 + \alpha + \frac{\beta_1^2}{E_1 - E_2}$$

$$\psi_1^{(1)} = \frac{H_{21}'}{E_1 - E_2} \cdot \psi_2^{(0)} = \frac{\beta_2}{E_1 - E_2} \cdot |\psi_2\rangle$$

$$\psi_2^{(1)} = \frac{\beta_1}{E_2 - E_1} |\psi_1\rangle$$

$$\psi_1' = |\psi_1\rangle + \frac{\beta_2}{E_1 - E_2} |\psi_2\rangle$$

$$\psi_2' = |\psi_2\rangle + \frac{\beta_1}{E_2 - E_1} |\psi_1\rangle$$